👩‍🏫 Staff Guide · Biology A · Science Starts: Safety & Lab Skills (Level 3)

Level 3 = resource-room students who can work close to the gen ed lesson. Same topics and order as Ms. Dunham's class; more reading and writing than Level 2.

1. How the pieces line up

Class lesson	Level 3 piece
Syllabus + FHS science safety contract	Reading part 1 + notes section A
Lollipop lab (observations, hypothesis, variables)	Reading parts 3-5 + notes sections B and C
A4 Hypotheses + Conclusion practice	Notes items on writing a hypothesis and conclusions; quick check

2. Supports built in

- Reading and notes each have a web copy with a 🔊 Read aloud button (highlights as it reads)
- Key-word box at the top of the notes; word problems have a worked example first
- Less filled in than Level 2: short answers, tables, a drawing
- No slide decks: students follow Ms. Dunham's slides with the Level 3 notes

3. Files

- Reading (+ 🔊 web copy)
- Guided notes + key (+ 🔊 web copy)
- Quick check + key
- Unit game: 10 questions

4. What to grade

- Guided notes stand in for the matching class notes or worksheet (same points when complete)
- Quick check: score it, then scale to the class quiz points
- Reading check questions: completion points
- Modified grading only as the IEP allows. Check with the case manager.

5. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Class content: Ms. Dunham.

🧪 Science Starts: Safety & Lab Skills: Guided Notes

Biology A · Unit 0 · Level 3 (closer to the class work) · goes with Start of Year: safety contract, Lollipop lab, A4 Hypotheses + Conclusion practice

Name: _____

Use the class slides and your reading. Key words:

goggles | *Code one* | *observation* | *inference* | *hypothesis* | *independent variable* | *dependent variable* | *data* | *conclusion*

A. Lab safety

1. Wear _____ any time there are chemicals, heat, or glass.
2. If someone is hurt, yell "_____!"
3. Report every _____ to the teacher, no matter how small.
4. Why is food not allowed in the lab?

B. Observations and hypotheses

5. An _____ uses your five senses.
6. A hypothesis is written as If… then… _____…
7. Write a hypothesis for this question: Does a lollipop dissolve faster in warm water or cold water?

C. Variables, tools and conclusions

8. The variable the scientist changes is the _____ variable.
9. Fill in the table.

Tool	What it measures	Unit
thermometer		
ruler		
graduated cylinder		
balance		

10. Your hypothesis was not supported. Did the experiment fail? Explain.

🧪 Science Starts: Safety & Lab Skills: Reading

Biology A · Unit 0 · Level 3 (closer to the class work) · goes with Start of Year: safety contract, Lollipop lab, A4 Hypotheses + Conclusion practice

Name: _____

How Scientists Work Safely

Every science lab starts with safety. At Ferndale, students sign the science safety contract. Goggles go on any time there are chemicals, heat, or glass. There is no food, drink, or gum in the lab. Long hair is tied back, shoes cover the whole foot, and loose sleeves and jewelry are secured.

Accidents must be reported to the teacher right away, even small ones. If someone is hurt, a student yells “Code one!” so the teacher comes fast. Following every direction, and asking when you are not sure, keeps everyone safe.

Scientists start with observations. An observation uses the five senses to describe what is really there: “The lollipop is red and round.” An inference is what you think it means: “It will taste like cherry.” Good science keeps the two apart.

Next, a scientist writes a hypothesis, a testable prediction. A strong hypothesis uses If… then… because…: “If I suck on the lollipop without biting, then it will dissolve in 8 minutes, because it is small.”

In an experiment, the independent variable is what the scientist changes. The dependent variable is what the scientist measures. Everything else stays the same. After collecting data, the scientist writes a conclusion: was the hypothesis supported or not supported, and what data shows it?

Check your reading

1. Name two things you must do to dress safely for lab.

_____

_____

2. What is the difference between an observation and an inference?

_____

_____

3. In the lollipop lab, what is the dependent variable?

_____

_____

🧪 Science Starts: Safety & Lab Skills: Quick Check

Biology A · Unit 0 · Level 3 (closer to the class work) · goes with Start of Year: safety contract, Lollipop lab, A4 Hypotheses + Conclusion practice

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. When must you wear goggles?

A. Any time there are chemicals, heat, or glass

B. Only when the teacher is watching

C. Only on Fridays

2. “The lollipop is red” is…

A. An inference

B. A hypothesis

C. An observation

3. A student changes the amount of light a plant gets. The amount of light is the…

A. Dependent variable

B. Conclusion

C. Independent variable

4. Which tool measures volume?

A. Balance

B. Graduated cylinder

C. Thermometer

5. Write a hypothesis using If… then… because… for: Do plants grow taller with more water?

6. What should you do first if your lab partner cuts their hand?

🧪 Science Starts: Safety & Lab Skills: Guided Notes · ANSWER KEY

Biology A · Unit 0 · Level 3 (closer to the class work) · goes with Start of Year: safety contract, Lollipop lab, A4 Hypotheses + Conclusion practice

Name: _____

Use the class slides and your reading. Key words:

goggles | *Code one* | *observation* | *inference* | *hypothesis* | *independent variable* | *dependent variable* | *data* | *conclusion*

A. Lab safety

1. Wear **goggles** any time there are chemicals, heat, or glass.
2. If someone is hurt, yell "**Code one**!"
3. Report every **accident** to the teacher, no matter how small.
4. Why is food not allowed in the lab?
Food could pick up chemicals or germs, and you could swallow something harmful.

B. Observations and hypotheses

5. An **observation** uses your five senses.
6. A hypothesis is written as If… then… **because** …
7. Write a hypothesis for this question: Does a lollipop dissolve faster in warm water or cold water?
Example: If the lollipop is in warm water, then it will dissolve faster, because heat speeds up dissolving.

C. Variables, tools and conclusions

8. The variable the scientist changes is the **independent** variable.
9. Fill in the table.

Tool	What it measures	Unit
thermometer	**temperature**	**°C**
ruler	**length**	**cm**
graduated cylinder	**volume**	**mL**
balance	**mass**	**g**

10. Your hypothesis was not supported. Did the experiment fail? Explain.
No. A hypothesis that is not supported still teaches you something; the data is the evidence.

🔑 *Key written by our team from Ms. Dunham's materials, please check.*

🧪 Science Starts: Safety & Lab Skills: Reading · ANSWER KEY

Biology A · Unit 0 · Level 3 (closer to the class work) · goes with Start of Year: safety contract, Lollipop lab, A4 Hypotheses + Conclusion practice

Name: _____

How Scientists Work Safely

Every science lab starts with safety. At Ferndale, students sign the science safety contract. Goggles go on any time there are chemicals, heat, or glass. There is no food, drink, or gum in the lab. Long hair is tied back, shoes cover the whole foot, and loose sleeves and jewelry are secured.

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Scientists start with observations. An observation uses the five senses to describe what is really there: “The lollipop is red and round.” An inference is what you think it means: “It will taste like cherry.” Good science keeps the two apart.

Next, a scientist writes a hypothesis, a testable prediction. A strong hypothesis uses If… then… because…: “If I suck on the lollipop without biting, then it will dissolve in 8 minutes, because it is small.”

In an experiment, the independent variable is what the scientist changes. The dependent variable is what the scientist measures. Everything else stays the same. After collecting data, the scientist writes a conclusion: was the hypothesis supported or not supported, and what data shows it?

Check your reading

1. Name two things you must do to dress safely for lab.
Any two: tie back long hair, wear closed-toe shoes, secure loose sleeves, take off dangling jewelry.

2. What is the difference between an observation and an inference?
An observation uses the five senses to describe what is there; an inference is what you think it means.

3. In the lollipop lab, what is the dependent variable?
The time it takes the lollipop to dissolve (what is measured).

🧪 Science Starts: Safety & Lab Skills: Quick Check · ANSWER KEY

Biology A · Unit 0 · Level 3 (closer to the class work) · goes with Start of Year: safety contract, Lollipop lab, A4 Hypotheses + Conclusion practice

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. When must you wear goggles?

A. Any time there are chemicals, heat, or glass | B. Only when the teacher is watching | C. Only on Fridays

2. "The lollipop is red" is…

A. An inference | B. A hypothesis | **C. An observation**

3. A student changes the amount of light a plant gets. The amount of light is the…

A. Dependent variable | B. Conclusion | **C. Independent variable**

4. Which tool measures volume?

A. Balance | **B. Graduated cylinder** | C. Thermometer

5. Write a hypothesis using If… then… because… for: Do plants grow taller with more water?

Example: If a plant gets more water, then it will grow taller, because plants need water to grow.

6. What should you do first if your lab partner cuts their hand?

Yell "Code one!" / tell the teacher right away.

🔑 *Key written by our team from Ms. Dunham's quizzes and notes, please check.*